

AEGIS

MEDIASUITE

User Manual — Version 2.1

The complete multimedia production suite for Linux

- Media Player ·
- DAW ·
- Mod Tracker ·
- Label Maker ·

- Audio Editor ·
- DJ Mixer ·
- Disc Tools ·
- Middleware ·

- Video Editor ·
- Karaoke ·
- Converter ·
- Launcher ·

Welcome to Aegis MediaSuite

Aegis MediaSuite is a comprehensive multimedia production environment for Linux. It bundles twelve specialised applications under a single unified launcher, sharing a common audio engine, library, and settings system. Whether you need to play back a playlist, edit audio waveforms, mix a DJ set, burn a disc, or produce a full multi-track recording, Aegis provides the tools in a consistent, keyboard-driven interface styled after KDE Breeze.

System Requirements

- Linux kernel 5.15 or later (X11 or Wayland)
- Qt 6.4+ with Quick, Qml, Multimedia, DBus, Svg modules
- MPV 0.35+ for media decoding
- FFmpeg 6.0+ (libavcodec, libavformat, libswresample)
- PipeWire or PulseAudio for audio output
- 4 GB RAM recommended (8 GB for DAW / Video Editor)
- OpenGL 3.3 or Vulkan 1.1 for hardware-accelerated video

Installation

Build from source using CMake 3.19+. After cloning the repository, run `cmake -B build -DCMAKE_BUILD_TYPE=Release` followed by `cmake --build build -j$(nproc)`. Install with `cmake --install build`. Each application is available as a symlink: `aegis-mediaplayer`, `aegis-audioeditor`, etc.

Running Individual Applications

Every symlink (e.g. `aegis-djmixer`) launches the main binary in the corresponding mode via `argv[0]` detection. You may also pass `--mode=<name>` explicitly, or open the unified Launcher and select the desired application tile.

Interface Conventions

- Dark theme throughout — optimised for low-light studio environments.
- Keyboard shortcuts follow standard conventions: Ctrl for file ops, Space for play/pause, arrow keys for navigation.
- All windows remember their size and position across sessions.
- Right-click context menus are available on most interactive elements.
- The status bar at the bottom always shows the current file, position, and level meters.

1 Launcher

The central hub for all Aegis applications and system settings.

1.1 Overview

The Launcher is the default entry point when running the bare **aegis** binary. It presents a grid of application tiles, each with its icon, name, and a brief description. Clicking a tile opens the corresponding application. Multiple applications can run simultaneously; they share the same audio backend process.

1.2 Application Grid

- Single-click a tile to launch the application.
- Right-click a tile to pin it to the taskbar or create a desktop shortcut.
- Drag tiles to rearrange them. The layout is saved automatically.
- The search bar (Ctrl+F) filters tiles by name or keyword.

1.3 Settings

The gear icon in the top-right opens the global Settings panel, which covers audio device selection, theme customisation, keyboard shortcut remapping, library paths, and plugin directories. Settings are shared across all applications and stored in `~/.config/Aegis/Aegis.conf`.

1.4 Keyboard Shortcuts

Shortcut	Action
Ctrl+F	Focus the application search bar
Enter	Launch the selected/focused application tile
Ctrl+,	Open global Settings
Ctrl+Q	Quit Aegis entirely
Alt+Tab	Switch between open Aegis windows

2 Media Player

Full-featured audio and video player with playlist management.

2.1 Overview

The Media Player is built on the MPV backend and supports virtually every audio and video format, including MP3, FLAC, OGG, AAC, OPUS, WAV, MP4, MKV, AVI, and WebM. It features a waveform visualiser, spectrum analyser, gapless playback, ReplayGain normalisation, and chapter navigation for long-form content.

2.2 Opening Files

- Drag files or folders from a file manager directly onto the player window.
- Use File → Open (Ctrl+O) to browse for individual files.
- Use File → Open Folder to enqueue all supported files in a directory.
- Pass file paths as command-line arguments to aegis-mediaplayer.
- The player integrates with MPRIS2, allowing control from any MPRIS-aware app.

2.3 Playlist

The playlist panel (toggle with P) shows the current queue. Tracks can be reordered by dragging. Right-clicking a track reveals options to remove it, show it in the file manager, or copy its path. The playlist is automatically saved on exit and restored on next launch. M3U and PLS playlist files are both supported for import and export.

2.4 Keyboard Shortcuts

Shortcut	Action
Space	Play / Pause
S	Stop
N / Right	Next track
P / Left	Previous track
Ctrl+O	Open file
Ctrl+L	Focus URL / path bar
F	Toggle fullscreen (video)
M	Mute / unmute
Up / Down	Volume +5% / -5%

[/]	Seek -10 s / +10 s
ctrl+[/]	Seek -60 s / +60 s
R	Cycle repeat mode (off / one / all)
Z	Toggle shuffle
I	Show track information panel

3 Audio Editor

Non-destructive waveform editor with effects chain and multi-format export.

3.1 Overview

The Audio Editor provides a traditional waveform view for editing single audio files. Operations are non-destructive: all edits are stored as an edit list and applied on export. The editor supports 32-bit float processing internally regardless of the source bit depth.

3.2 Waveform View

- Scroll horizontally to navigate; use Ctrl+Scroll to zoom in/out.
- Click to set the playback cursor; click-drag to create a selection.
- The ruler at the top shows time in HH:MM:SS.ms or samples (toggle with T).
- Markers can be placed with M; named markers appear in the markers panel.
- The overview strip at the bottom shows the entire file for quick navigation.

3.3 Editing Operations

- Cut (Ctrl+X): removes the selected region and places it on the clipboard.
- Copy (Ctrl+C): copies the selected region.
- Paste (Ctrl+V): inserts clipboard audio at the cursor position.
- Delete: silences the selected region without shifting subsequent audio.
- Trim to Selection (Ctrl+T): discards everything outside the selection.
- Fade In / Fade Out: applies a linear or logarithmic fade to the selection.
- Normalize: scales the selection to a target dBFS peak.
- Silence: fills the selection with digital silence.

3.4 Effects

Open the Effects panel with E. Effects can be added from the built-in library (equaliser, compressor, reverb, noise gate, limiter, pitch shift, time stretch) or from installed LADSPA and LV2 plugins. Effects are applied non-destructively to the selection or the whole file.

3.5 Export

Export via File → Export (Ctrl+Shift+E). Supported formats include WAV, FLAC, MP3 (via LAME), OGG Vorbis, OPUS, AAC, and AIFF. Batch export is available from the Converter application.

3.6 Keyboard Shortcuts

Shortcut	Action
Space	Play / Pause from cursor
Ctrl+Z	Undo
Ctrl+Shift+Z	Redo
Ctrl+A	Select all
Ctrl+D	Deselect
Ctrl+X/C/V	Cut / Copy / Paste
Delete	Silence selection
Ctrl+T	Trim to selection
M	Add marker at cursor
E	Toggle Effects panel
Ctrl+Shift+E	Export file
+ / -	Zoom in / out
Home / End	Go to start / end

4 Video Editor

Timeline-based video editor with multi-track audio and real-time preview.

4.1 Overview

The Video Editor provides a multi-track timeline for assembling video projects. It uses FFmpeg for decoding and re-encoding, and MPV for real-time preview. Video effects are applied via a node-based effects graph accessible from the Effects panel. Supported output formats include MP4 (H.264/H.265), MKV, WebM, and lossless intermediates.

4.2 Project Structure

- The Media Pool (left panel) holds all imported clips.
- The Timeline (centre) contains video and audio tracks arranged in time.
- The Preview Monitor (top-right) shows the frame at the playhead position.
- The Properties panel (right) shows clip or effect parameters for the selection.
- Projects are saved in the native .aegisproj format (XML-based).

4.3 Timeline Editing

- Drag clips from the Media Pool to the timeline to add them.
- Trim a clip by dragging its left or right edge.
- Split a clip at the playhead with Ctrl+K.
- Ripple-delete a gap with Shift+Delete.
- Transitions (cross-dissolve, wipe, etc.) are added by overlapping two clips.
- Multiple audio tracks can be mixed; volume automation is drawn directly on the clip.

4.4 Export

Use File → Export to render the project. The export dialog allows selection of codec, resolution, frame rate, bitrate preset, and audio settings. A background rendering queue lets you continue editing while exporting.

4.5 Keyboard Shortcuts

Shortcut	Action
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Space	Play / Pause preview
Ctrl+K	Split clip at playhead
Ctrl+Z / Y	Undo / Redo
I / O	Set In / Out point
Shift+Delete	Ripple delete selection
Ctrl+R	Render export queue
Ctrl+Shift+I	Import media into pool
G	Toggle snap to grid
, / .	Step one frame back / forward

5 DAW (Digital Audio Workstation)

Multi-track recording, MIDI sequencing, and mixing with VST/LV2 plugin support.

5.1 Overview

The DAW is the centrepiece of Aegis for music production. It supports unlimited audio and MIDI tracks (hardware permitting), a built-in mixer with channel strips, send/return routing, and a master bus. Audio is processed at the project sample rate (44.1–192 kHz) and bit depth (16/24/32-bit float). MIDI is routed via ALSA Sequencer.

5.2 Tracks

- Audio tracks record and play back audio clips.
- MIDI tracks drive software instruments (built-in synthesiser or VST/LV2).
- Instrument tracks combine a MIDI sequencer with a built-in instrument.
- Bus tracks aggregate sends from multiple channels for group processing.
- The master track controls overall output volume and applies mastering effects.

5.3 Recording

Arm a track for recording by clicking its REC button. Set the input device and channel in the track header. Press R (global record) then Space to start recording. Punch-in/out is supported by setting In/Out markers. Overdub mode layers new recordings on top of existing clips.

5.4 Mixer

The Mixer (Ctrl+M) shows a channel strip for every track: fader, pan knob, solo/mute buttons, pre/post-fader send levels, and an insert effects chain. Effects are loaded by clicking the FX slot — VST2, VST3, LADSPA, and LV2 plugins are scanned from the paths configured in Settings → Plugins.

5.5 MIDI Sequencer

Double-click a MIDI clip to open the Piano Roll. Notes are drawn, resized, and moved with the mouse. Velocity is shown as bar height. The Quantize panel aligns notes to a selectable grid. Chord, arpeggio, and humanise tools are available in the Edit menu.

5.6 Keyboard Shortcuts

Shortcut	Action
Space	Play / Stop
R	Toggle global record
Ctrl+M	Toggle Mixer panel
Ctrl+P	Open Piano Roll for selected MIDI clip
Ctrl+D	Duplicate selected clip
Ctrl+G	Group selected clips
T	Select tool (pointer)
A	Select draw / pencil tool
S	Select split tool
Ctrl+Shift+Q	Quantize selection
Ctrl+R	Toggle loop region
F5	Bounce / render selection to audio

6 DJ Mixer

Two-deck DJ mixer with BPM detection, beatmatching, and EQ/FX per deck.

6.1 Overview

The DJ Mixer presents two virtual decks (A and B) and a central mixer section. Each deck shows a scrolling waveform, BPM counter, cue point buttons, and loop controls. The crossfader blends between the decks. An effects unit with reverb, echo, flanger, and filter is available per deck.

6.2 Loading Tracks

- Drag a track from the integrated library browser to either deck.
- Click the folder icon on a deck to browse for a file.
- Recently played tracks appear in the history pane at the bottom.
- BPM is detected automatically on load; it can be tapped or set manually.

6.3 Beatmatching

Use the pitch slider on each deck to nudge the BPM. The Sync button automatically matches the tempo and phase of the non-master deck to the master. The waveform overlay view (press W) shows both decks' waveforms superimposed to help manual alignment.

6.4 Cue Points and Loops

- Set up to eight hot cue points per track with the HOT CUE buttons.
- Press a hot cue to jump to that position instantly.
- Mark a loop start with IN and loop end with OUT; the loop plays until cancelled.
- Loop size can be halved or doubled with the /2 and ×2 buttons.

6.5 Keyboard Shortcuts

Shortcut	Action
Q	Play / Pause Deck A
W	Cue Deck A (play from cue, release to return)
O	Play / Pause Deck B

P	Cue Deck B
1–8	Trigger hot cue 1–8 on active deck
Ctrl+1–8	Set hot cue 1–8 on active deck
F1 / F2	Deck A tempo –0.5% / +0.5%
F9 / F10	Deck B tempo –0.5% / +0.5%
Tab	Switch active deck focus
S	Toggle Sync on active deck

7 Karaoke

Karaoke player with lyrics display, pitch correction, and microphone mixing.

7.1 Overview

The Karaoke application plays CDG+MP3, LRC, and KAR files. Lyrics scroll in sync with the music and are highlighted word-by-word. The built-in vocal processor applies reverb and optional pitch correction to the microphone input. A dual-display mode sends the lyrics to a second monitor or projector.

7.2 Loading Songs

- Open a song queue (Ctrl+O) or drag files onto the application.
- Supported formats: MP3+CDG pairs, LRC sidecar files, KAR MIDI karaoke.
- The song browser searches your library by title, artist, or language tag.
- Songs are queued; the next song loads automatically at the end of each track.

7.3 Vocal Controls

- Microphone volume is controlled with the MIC knob.
- Key shift transposes the backing track up or down in semitones (± 6).
- Reverb wet/dry is adjustable in real time.
- Pitch correction strength is set from 0 (off) to 100 (strong correction).

7.4 Display Settings

In Settings → Display, choose the font, size, and colours for the lyrics. The secondary display can be configured as a mirror or extended view. A countdown timer before each song gives singers time to prepare.

8 Mod Tracker

Classic tracker interface for composing MOD, XM, IT, and S3M modules.

8.1 Overview

The Mod Tracker is a step-sequencer inspired by the Amiga demoscene tradition. Music is organised into patterns, each containing note/instrument/effect data for up to 64 channels. Patterns are arranged in a song order list. Playback uses the libopenmpt engine for accurate format reproduction.

8.2 Pattern Editor

- Each row represents one tick (1/rows_per_beat of a beat).
- Enter notes by typing on the computer keyboard (QWERTY piano layout).
- Instrument number, volume, and effect columns follow each note cell.
- Effects use the standard MOD effect command syntax (e.g., Axx = set tempo).
- Multiple patterns can be selected and operated on as a block.

8.3 Samples and Instruments

Samples (raw audio data) are loaded from WAV, IFF, or embedded in module files. Instruments layer multiple samples with note ranges, envelopes, and vibrato settings. The built-in sample editor provides basic loop point and fade editing.

8.4 Keyboard Shortcuts

Shortcut	Action
Space	Play / Stop from current pattern
F5	Play song from start
F6	Play pattern looped
Ins	Insert row
Del	Delete note at cursor
Ctrl+I	Interpolate values in selection
Alt+D	Double pattern length
Alt+H	Halve pattern length
Ctrl+Up/Down	Move pattern order up / down

9 Disc Tools

CD/DVD/Blu-ray burning, ripping, ISO creation, and disc information.

9.1 Overview

Disc Tools consolidates all optical-disc operations: ripping audio CDs to FLAC/MP3/WAV, burning audio or data CDs and DVDs, creating and mounting ISO images, and reading disc metadata via MusicBrainz. It uses libcdio for ripping and libburn/libisofs for burning.

9.2 Audio CD Ripping

- Insert a disc and click Rip CD; the disc is read and tracks listed automatically.
- Metadata is fetched from MusicBrainz; edit fields manually if needed.
- Select output format (FLAC recommended), bitrate (for lossy formats), and destination folder.
- C2 error correction and AccurateRip verification are available in Settings.
- Ripping runs in the background; progress is shown in the Jobs panel.

9.3 Burning

- Drag files or an ISO image to the Burn queue.
- Select disc type (CD/DVD/BD-R), speed, and write mode (disc-at-once recommended).
- Audio CD mode arranges tracks with correct 2-second gaps and writes RedBook-compliant audio.
- Data disc mode creates a Joliet+Rock Ridge filesystem automatically.
- Verify after burn re-reads the disc and checks MD5 hashes of all written files.

9.4 ISO Operations

- Create ISO: select a folder or disc content, set the volume label, and save.
- Extract ISO: mount virtually or extract all files to a directory.
- Mount ISO: creates a temporary FUSE mount point (no root required).

10 Converter

Batch audio and video format converter with preset management.

10.1 Overview

The Converter provides a simple drag-and-drop interface for converting audio and video files between formats. Conversions run concurrently up to a configurable number of parallel jobs, using FFmpeg under the hood. Presets can be saved and shared as JSON files.

10.2 Adding Files

- Drag files or entire folders onto the file list.
- Use Add Files (Ctrl+O) or Add Folder (Ctrl+Shift+O) from the toolbar.
- Files can be reordered; only selected files are converted if a selection exists.
- Duplicate entries are detected and highlighted.

10.3 Output Settings

- Audio: format (WAV, FLAC, MP3, OGG, OPUS, AAC, AIFF), sample rate, bit depth/bitrate.
- Video: container (MP4, MKV, WebM), video codec (H.264, H.265, VP9, AV1), CRF, audio codec.
- Output directory: same as source, custom folder, or alongside source with suffix.
- Filename template supports variables: {title}, {artist}, {track}, {year}, {ext}.

10.4 Presets

Built-in presets cover common targets: MP3 320k, FLAC lossless, OGG high quality, AAC for iOS, H.264 1080p30, and more. Custom presets are saved to `~/.config/Aegis/converter_presets/`.

11 Label Maker

Design and print CD/DVD labels and cover inserts.

11.1 Overview

Label Maker is a lightweight design tool for creating printable CD/DVD labels, jewel case inserts, and booklets. It includes templates for standard disc sizes (74mm/120mm), slim jewel case, and digipak. Designs are exported to PDF for printing on standard label sheets.

11.2 Canvas and Templates

- Choose a template from the gallery (File → New from Template).
- The canvas is a true-to-scale representation of the label/insert.
- Ruler guides and snap-to-grid assist precise placement.
- Template layers (background, artwork, text, barcode) are individually shown/hidden.

11.3 Adding Content

- Text: click to place a text box, type content, set font/size/colour.
- Images: drag image files onto the canvas; resize and rotate with handles.
- Track listing: import from a CUE sheet or type manually in the track list editor.
- Barcode: auto-generates an EAN-13 barcode from a catalogue number.
- Background: solid colour, gradient, or full-bleed image fill.

11.4 Export and Print

Export to PDF (File → Export PDF, Ctrl+Shift+E) at 300 DPI for print-ready output. The print dialog maps the design to the correct label sheet template (Avery, Zweckform, etc.) and handles margins automatically.

12 Middleware

Audio routing, device management, and inter-application connectivity.

12.1 Overview

Middleware is the infrastructure layer of Aegis. It provides a visual patchbay for PipeWire/PulseAudio connections, MIDI device routing, virtual cable management, and system audio monitoring. It is also the host for the JACK compatibility layer used by professional audio plugins.

12.2 Audio Patchbay

- Drag a port from a source node to a sink node to create a connection.
- Connections appear as coloured lines; click a line to remove it.
- Nodes represent hardware devices, Aegis applications, and PipeWire virtual nodes.
- Saved patch configurations can be restored automatically on startup.
- JACK mode exposes all Aegis audio I/O as JACK ports for use with JACK-native software.

12.3 MIDI Routing

The MIDI patchbay tab shows all ALSA MIDI clients. Virtual MIDI cables connect controllers to instruments. Filters can block specific channels or remap controller numbers before forwarding. Clock sync (MIDI clock and MTC) can be generated or consumed from any port.

12.4 System Monitor

The Monitor tab displays real-time CPU, DSP load, audio dropout counter, buffer fill percentage, and active connection count. Alerts are triggered when the DSP load exceeds a configurable threshold, helping identify latency or CPU performance issues before they cause audible artefacts.

Appendix A — Global Keyboard Shortcuts

The following shortcuts work in any Aegis application window.

Shortcut	Action
Ctrl+Q	Quit the current application
Ctrl+W	Close the current window / return to Launcher
Ctrl+,	Open Settings for the current application
F1	Open context-sensitive Help
Ctrl+Shift+L	Open Library browser
Ctrl+Shift+M	Show/hide Middleware patchbay
Ctrl+Shift+P	Show/hide global Performance monitor
Alt+F4	Close window (standard)

Appendix B — Supported File Formats

Category	Formats
Audio playback	MP3, FLAC, OGG Vorbis, OPUS, AAC, WAV, AIFF, WMA, APE, TTA, WavPack
Video playback	MP4, MKV, AVI, MOV, WebM, TS, FLV, WMV, OGV
Audio export	WAV (16/24/32-bit), FLAC, MP3, OGG, OPUS, AAC, AIFF
Video export	MP4 (H.264/H.265), MKV, WebM (VP9/AV1)
Module tracker	MOD, XM, IT, S3M, MED, OKT, 669, FAR, MTM
Karaoke	MP3+CDG, LRC, KAR, MIDI
Disc images	ISO 9660, UDF, BIN/CUE
Projects	.aegisproj (XML), .aup3 (Audacity compat.), DAWPROJECT

Appendix C — Configuration Files

All configuration is stored under `~/.config/Aegis/`:

- **Aegis.conf** — global settings (Qt INI format)
- **library.db** — SQLite media library
- **keybindings.conf** — custom keyboard shortcuts
- **converter_presets/** — saved converter presets (JSON)
- **patches/** — saved Middleware patchbay configurations
- **themes/** — custom colour themes (JSON)

Appendix D — Getting Help

- Press F1 in any application for context-sensitive help.
- Project homepage and issue tracker: <https://github.com/aegis-mediasuite>
- Community forum and chat: see the project homepage for current links.
- Report bugs with the output of `aegis --version` and the relevant log from `~/local/share/Aegis/aegis.log`.